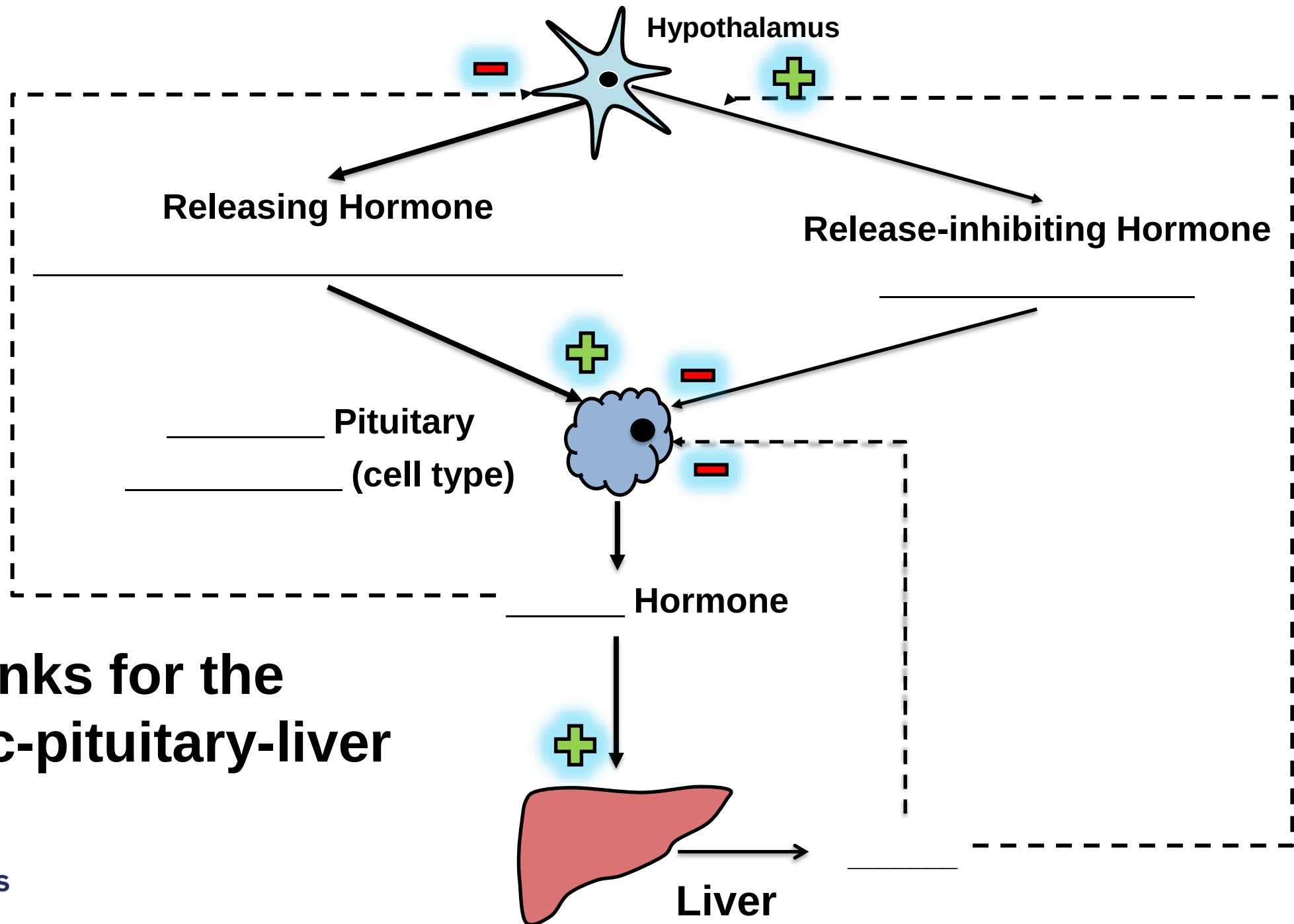


Growth Hormone Physiology

Objectives

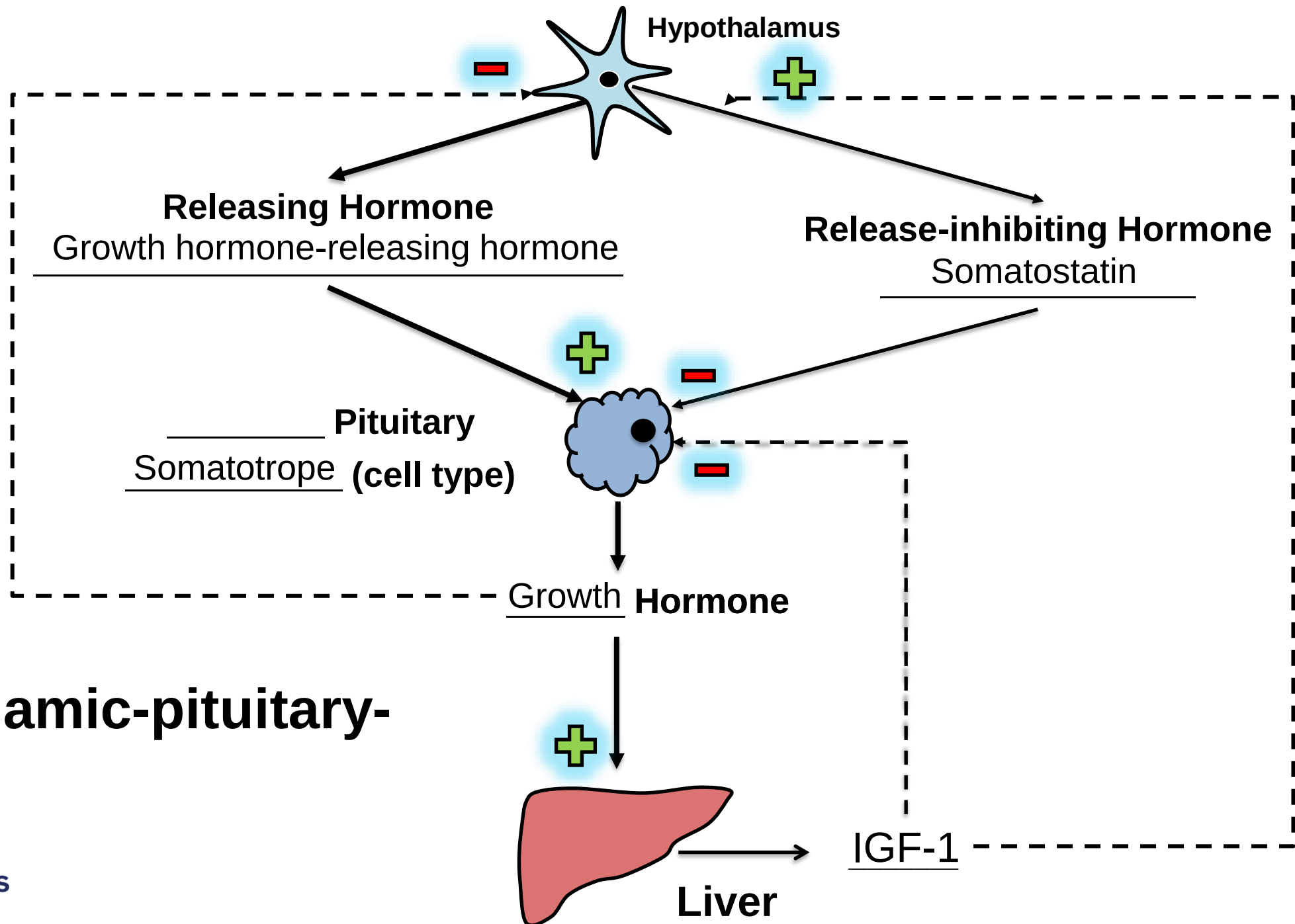
1. Describe the regulation of growth hormone secretion, to include the roles of hypothalamic factors, glucose, and IGF-I.
2. Identify the target organs or cell types for insulin like growth factors that account for longitudinal growth.
3. Describe the relationship between growth hormone and the insulin-like growth factors and their plasma binding proteins in the regulation of growth.
4. Specify the etiologies of gigantism, acromegaly, and Laron syndrome.
5. Describe the metabolic and growth promoting actions of growth hormone.

Describe the regulation of growth hormone secretion, and identify the roles of hypothalamic factors, glucose, and IGF-1.



Fill in the blanks for the hypothalamic-pituitary-liver axis!

Describe the regulation of growth hormone secretion, and identify the roles of hypothalamic factors, glucose, and IGF-1.



The hypothalamic-pituitary-liver axis

An infant is brought to his pediatrician by his mother, who says he is fussy and continually wanting to feed. The physician says that he is likely going through a growth spurt. Secretion of which hormone is likely involved in this situation?

- A. ACTH
- B. Epinephrine
- C. Ghrelin
- D. GnRH
- E. Vasoactive Intestinal Polypeptide

Answer: An infant is brought to his pediatrician by his mother, who says he is fussy and continually wanting to feed. The physician says that he is likely going through a growth spurt. Secretion of which hormone is likely involved in this situation?

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Ghrelin is a 'hunger hormone' produced by the stomach but is also expressed in the hypothalamus. It stimulates GH secretion by action on somatotropes. Epinephrine can stimulate GH secretion but in a growth spurt this is not the likely mechanism. ACTH, GnRH and VIP have no effect on GH secretion.

Which of the following stages of life is associated with the highest growth hormone secretion?

- A. Birth
- B. Childhood
- C. Puberty
- D. Adulthood (<65 years)
- E. Elderly

Answer: Which of the following stages of life is associated with the highest growth hormone secretion?

- A. Birth
- B. Childhood
- C. Puberty**
- D. Adulthood (<65 years)
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Puberty is associated with a large growth spurt. This is associated with the highest rates of GH secretion, due to stimulation by estrogen and testosterone. In males, estrogen rises through conversion of testosterone by aromatase.

A 43-year-old male has some labs drawn as part of their annual check up. The labs reveal elevated serum glucose and elevated serum free fatty acid levels. What would you expect about the level of GH secretion in this patient?

- A. Decreased
- B. Increased
- C. Normal

Answer: A 43-year-old male has some labs drawn as part of their annual check up. The labs reveal elevated serum glucose and elevated serum free fatty acid levels. What would you expect about the level of GH secretion in this patient?

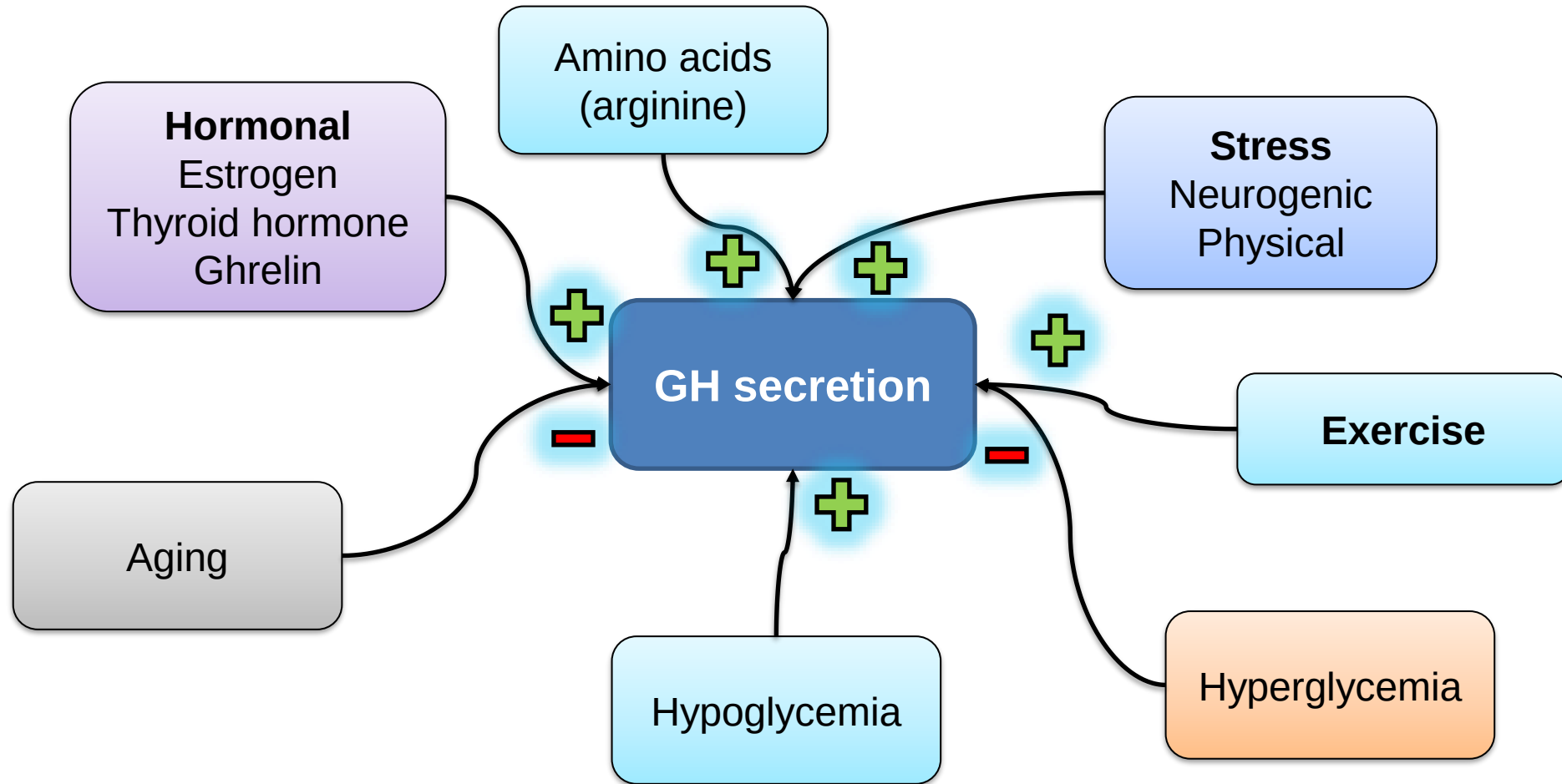
A. Decreased

B. Increased

C. Normal

Hyperglycemia and increased serum FFA inhibit GH secretion.

Summary of factors that regulate GH secretion



A 7-year-old male comes to your clinic for his yearly check up. He was in the 5th percentile for height until the age of 4 but has now fallen to below the 3rd percentile.

You decide you want to assess the function of the hormonal control of growth in the patient.

To clinically assess the function of the hormonal regulation of growth, measurement of the serum levels of which of the following would be best?

- A. Growth hormone
- B. Growth hormone-releasing hormone
- C. Insulin-like Growth Factor-1 (IGF-1)
- D. Somatostatin
- E. Thyroid hormone

Answer: To clinically assess the function of the hormonal regulation of growth, measurement of the serum levels of which of the following would be best?

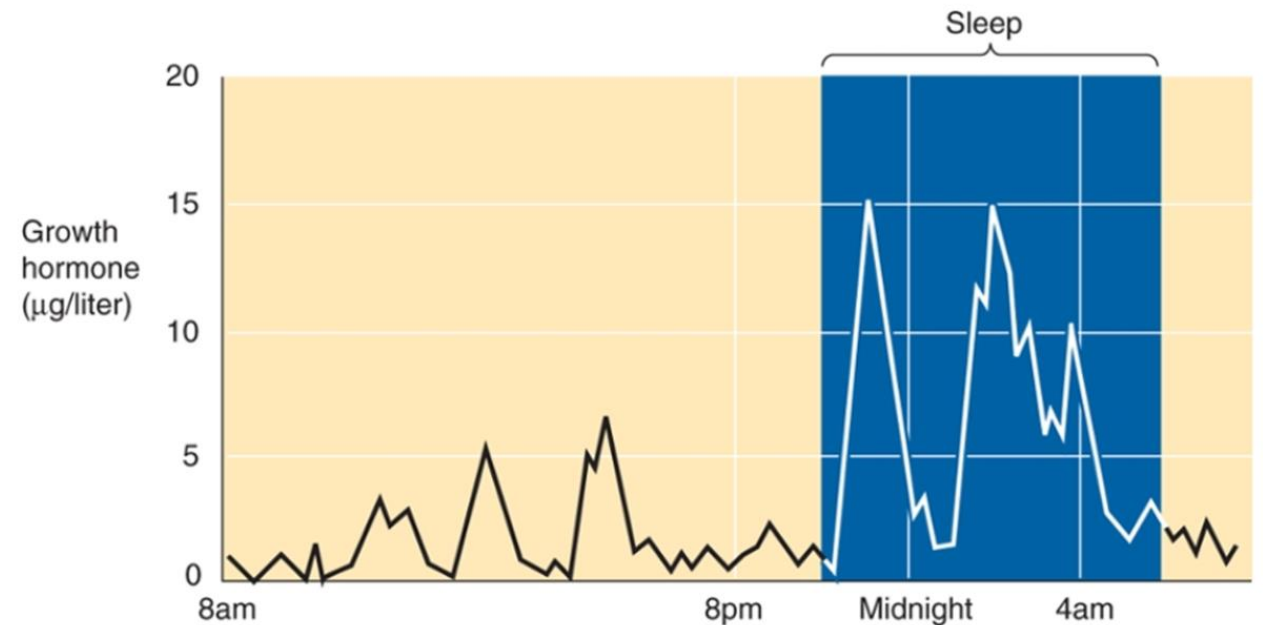
- A. Growth hormone
- B. Growth hormone-releasing hormone
- C. Insulin-like Growth Factor-1 (IGF-1)**
- D. Somatostatin
- E. Thyroid hormone

The next slide explains why there is limited clinical utility in measuring serum GH.

Limited clinical utility of measuring serum GH levels

Marked variation in serum GH levels:

- Pulsatile secretion
- Short half life
- Diurnal rhythm of secretion
 - Peak secretion in early morning before awakening
- Lowest secretion during the day



Which of the following are functions of IGF-1 binding protein? Select all that apply

- A. Diminish the activity of IGF-1
- B. Inhibit GHRH secretion
- C. Increase diffusion of IGF-1 into cells
- D. Increase efficacy at the IGF-1 receptor
- E. Prolong the half-life of IGF-1

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Function of IGF-BPs:

- Carry IGF-I in the circulation
- Inhibit the actions of IGF-1 (prevents systemic activity)
- Greatly extends half-life of IGF-I

Prostate cancer cells produce IGF-binding protein proteases. Why?



Answer: Prostate cancer cells produce IGF-binding protein proteases. Why?

- Degradation of IGF binding protein liberates locally free IGF
- Stimulates cancerous growth in the prostate gland selectively

In bone, which of the following is most important for increasing linear growth?

- A. Growth hormone
- B. IGF-BP
- C. IGF-1
- D. IGF-2

Answer: In bone, which of the following is most important for increasing linear growth?

A. Growth hormone

B. IGF-BP

C. IGF-1

D. IGF-2

GH does have some direct action on bone, but most actions of GH are mediated through IGF-1 (Somatomedin C).

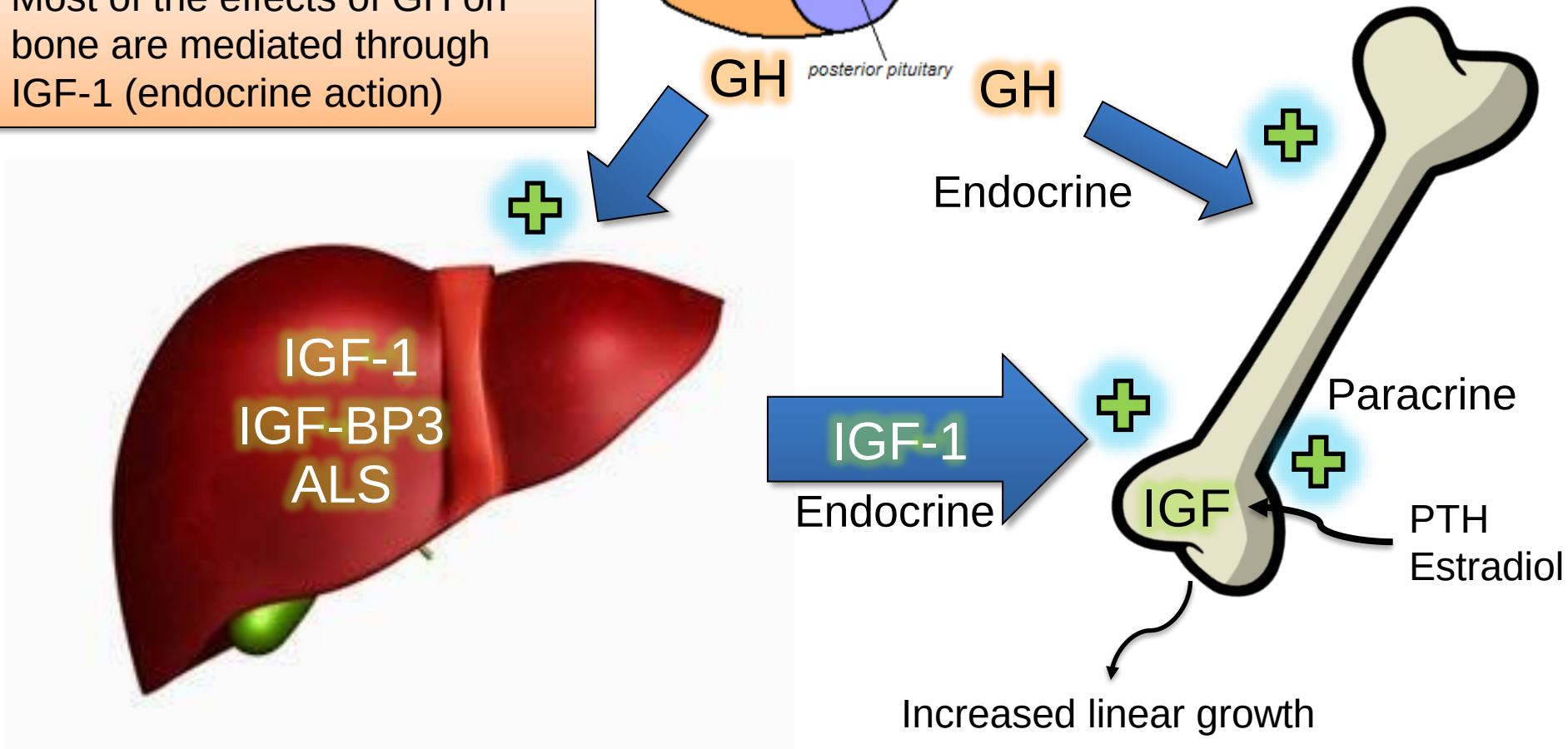
IGF-2 is the major form produced in the fetus, where it regulates growth and development in a GH-independent manner.

IGF-BP carries GH in the circulation but is otherwise biologically 'inert'.

Describe the relationship between growth hormone and the insulin like growth factors and their plasma binding proteins in the regulation of growth.

Identify the target organs or cell types for insulin like growth factors that account for longitudinal growth.

Most of the effects of GH on bone are mediated through IGF-1 (endocrine action)



GH does have some direct effects on bone independent of IGF-1.

IGF-1 is also produced locally in bone without the involvement GH, coming from local paracrine actions or from other hormones such as PTH or estradiol

Describe the metabolic and growth promoting actions of growth hormone.

A 7-year-old female is referred to an endocrinologist for below normal growth. Labs show a normal GH release in response to exercise, but low levels of serum IGF-1. Which of the following is the likely mechanism underlying her lack of growth?

- A. Decreased GHRH secretion
- B. Decreased GH receptor expression on bone
- C. Decreased GH receptor sensitivity
- D. Decreased GH synthesis in the anterior pituitary

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The low levels of IGF-1 indicate a problem with IGF-1 production in response to GH release. Since GH release is normal in response to exercise, GH synthesis and GHRH secretion must be normal. Therefore, the likely mechanism is diminished GH receptor sensitivity, meaning that less IGF-1 is produced in response to GH in the liver. GH's effects on bone are mostly mediated through IGF-1, so decreased GH receptors on bone cannot be the mechanism.

Growth abnormalities due to GH excess or deficiency

- Before adulthood:
 - Growth impairment
 - Isolated GH deficiency (mutations in GH/GHRH genes)
 - Laron Syndrome: GH resistance due to mutation of the GH receptor
 - Gigantism
 - Excess GH
- After adulthood (closure of epiphyses):
 - No impairment of longitudinal growth
 - Excess GH causes acromegaly (appositional growth)

Which of the following explains why individuals who strength train should ingest protein after working out?

- A. GH stimulates amino acid uptake in muscle
- B. It decreases insulin secretion
- C. It tastes good
- D. Protein stimulates lipolysis
- E. Protein decreases metabolic rate

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- D. Protein stimulates lipolysis
- E. Protein decreases metabolic rate

GH secretion is stimulated by exercise and increases amino acid uptake. In the presence of adequate protein, protein synthesis and muscle growth is stimulated by GH (this is dependent on existence of the Fed state). Lipolysis occurs in the fasted state. Protein consumption increases metabolic rate and also stimulates insulin secretion. It does taste good though (arguably).

A 12-year-old male is visiting his pediatrician for a physical. The pediatrician asks him to describe his diet. He lists white bread, cookies and fruit juice as his staples. What effect will GH have with this diet?

- A. Caloric storage
- B. Decreased adiposity
- C. Increased growth
- D. Lipid mobilization
- E. Protein synthesis

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The lack of dietary protein/amino acids as well as elevated serum glucose and insulin levels will inhibit GH & IGF secretion. This favors caloric storage (fat deposition) rather than linear growth, protein synthesis or lipid mobilization.

A 41-year-old physiology professor goes on a crash diet to lose weight. He does not consume any food for two days. What is the effect of growth hormone in this circumstance?

- A. Decreases gluconeogenesis
- B. Increases glucose uptake into muscle
- C. Increases lipolysis
- D. Increases protein synthesis

Answer: A 41-year-old physiology professor goes on a crash diet to lose weight. He does not consume any food for two days. What is the effect of growth hormone in this circumstance?

- A. Decreases gluconeogenesis
- B. Increases glucose uptake into muscle
- C. Increases lipolysis**
- D. Increases protein synthesis

In the fasted state, GH has a lipolytic effect on adipose tissue. There is decreased glucose uptake into adipose tissue and muscle. Protein synthesis would be decreased in the fasted state. Gluconeogenesis in the liver is stimulated by GH, not decreased, in the fasted state.

A patient with acromegaly is noted to have high fasting serum glucose levels. Which of the following mechanisms contributes to this?

- A. Decreased gluconeogenesis
- B. Decreased insulin secretion
- C. Decreased insulin sensitivity
- D. Increased glucagon secretion
- E. Increased glycogenolysis

Answer: A patient with acromegaly is noted to have high fasting serum glucose levels. Which of the following mechanisms contributes to this?

- A. Decreased gluconeogenesis
- B. Decreased insulin secretion
- C. Decreased insulin sensitivity**
- D. Increased glucagon secretion
- E. Increased glycogenolysis

GH is considered a diabetogenic hormone: Mechanisms involved in GH-mediated hyperglycemia

1. Decreased glucose uptake into muscle and adipose tissue
 - due to increased serum FFA from lipolysis
2. Stimulation of liver gluconeogenesis
 - Increased liver acetyl CoA from fatty acid oxidation
 - GH-mediated increase in PEPCK expression
3. Insulin insensitivity
 - GH antagonizes the action of insulin at the post-receptor level in skeletal muscle & adipose tissue